



Unit 02: Project Administration

Topics to be covered:

- 2.1 Essentials of Project Administration
- 2.2 Project Team
- 2.3 Project Design
- 2.4 Work Breakdown Structure (WBS)
- 2.5 Project Execution Plan (PEP)
- 2.6 Contracting Plan
- 2.7 Work Packing Plan
- 2.8 Organization Plan
- 2.9 Systems and Procedure Plan
- 2.10 Project Procedure Manual
- 2.11 Project Diary
- 2.12 Project Execution System – List its forms
- 2.13 List the Pre-requisites for Successful Project Implementation
- 2.14 Case Studies

2.1 Essentials of Project Administration:

- ☐ Project administration focus on the effective:
 - a) Project initiation
 - b) Project Planning
 - c) Project execution
 - d) Project Monitoring and
 - e) Project Closure.
- ☐ The process of providing a project management service to the project teams is called project administration.
- ☐ Project administration assist a project manager by performing many of the administrative tasks required for the smooth running of the project.

a) Project Initiation:

- ☐ Project initiation is the first phase (stage) in managing a project.
- ☐ It sets the foundation for everything that comes next.
- ☐ It ensures the project is clear, realistic and ready to begin.

Key Objectives of Project Initiation:

- Understand what the project is about and why it is important.
- Check if the project is possible with the available resources, time and budget.
- Identify the people (stakeholders) involved and their roles.
- Get official approval to start the project.

b) Project Planning:

- ☐ Project planning is the second phase (stage) of project management.
- ☐ It involves creating a roadmap to guide the team on how to complete the project successfully.
- ☐ This phase ensures clarity, organization and preparation before work begins.

Key Objectives of Project Planning:

- Break the project into smaller, manageable tasks.
- Set clear timelines and deadlines.
- Allocate resources (people, money, tools) efficiently.
- Identify risks and prepare solutions.
- Establish communication plan (how communication will happen).

c) Project Execution:

- ☐ Project execution is the phase where the work is done as per the plan.
- ☐ It involves coordinating people, resources and tasks to complete project goals.
- ☐ This is the "action" phase of project management.

Key Objectives of Project Execution:

- Complete tasks and deliverables as per the plan.
- Ensure the team stays on fixed time and within budget.
- Maintain communication and address issues (problems) as they arise.
- Monitor progress to ensure quality and efficiency.

d) Project Monitoring:

- ☐ Project monitoring is the phase where you track the progress and performance of the project to ensure it stays on track.
- ☐ It helps identify issues (problems) early so corrective actions can be taken.
- ☐ It ensures project goals are achieved, within the planned timeline, budget and scope.

Key Objectives of Project Monitoring:

- Measure progress against the project plan.
- Ensure tasks are completed on time and within budget.
- Identify and resolve issues or risks early.
- Maintain quality standards.

e) Project Closure:

- ☐ Project closure is the final phase of project management.
- ☐ It involves wrapping up all project activities, delivering the final product or service and ensuring all stakeholders are satisfied.
- ☐ This phase ensures the project is formally completed and lessons are captured for future projects.

Key Objectives of Project Closure:

- Deliver the project's final results to the client or stakeholders.
- Confirm all project tasks are completed.
- Release project resources (team members, tools and budgets).
- Document lessons learned and best practices.
- Celebrate the team's success!

Q. Define project administration and list any 3 tools used in project administration.

- ☐ The process of providing a project management service to the project teams is called **Project Administration**.

The 3 tools used in project administration are as follows:

- i. Work Breakdown Structure (WBS)
- ii. Project Execution Plan (PEP)
- iii. Project Procedure Manual (PPM).

2.2 Project Team:

- ☐ A project team is a group of individuals (team members or project staff) who have specific roles in the project.
- ☐ Project team consists of members with different skills and knowledge.
- ☐ Each team members are assigned with specific roles and responsibilities that contribute to the project's success.

Project team can be classified as:

- i. Initial project team
- ii. Project leader/manager
- iii. Core project team
- iv. Full project team
- v. Project advisors
- vi. Project stakeholders
- vii. Project or Process facilitators.

i) Initial project team

- It consists of specific people who have idea of starting a project.
- The member of this team may or may not be a part of the core project team.

ii) Project leader/manager

- Any one of the team members will be designated as a project leader/manager based on knowledge and experience and he/she will be responsible for the completion of project.

iii) Core project team

- Small group of people of 3 to 8 members who are ultimately responsible for designing and managing the project are called core project team.
- Example: Investor, customer, manager, consultants and internal auditor (inspector or examiner).

iv) Full Project Team

- Group of people involved in designing, implementing, monitoring and controlling the projects are called full project team.

v) Project Advisors

- A person who provides expert advice to the project team is called project advisors.
- A project advisor is not the part of project team but provide only the expert advice.

vi) Project stakeholders

- People having the stake (amount of ownership) in the project are called project stake holders.
- They are interest in the project outcome.

vii) Project Facilitators

- Project Facilitator help the project through the planning process.
- He is part of the initial project team and the core project team.
- He understands the key elements of the project processes.
- He has good communication skills.
- A facilitator is an unbiased person.
- The facilitator will solve problems by listening to both sides of an argument.

Advantages of effective team.

Following are the advantages of ineffective team:

- ☐ Clear objective (goals) of the project from the start to finish.
- ☐ Good decision making, which will speed up the project activities.
- ☐ Clear roles, responsibilities and leadership qualities.
- ☐ Ensures smooth progress of the project without any delay.
- ☐ Leadership roles are shared by the team leader.
- ☐ Trust, co-operation, support and constructive conflict or feedback.
- ☐ Individual and mutual support for end results.

Pitfalls or disadvantages of ineffective team.

Following are the disadvantages of ineffective team:

- ☐ Low level of motivation in the project team.
- ☐ Confusion about responsibility.
- ☐ Poor communication from project manager.
- ☐ Poor communication among the team members.
- ☐ Conflicts (fights) among team members.

- ☐ Slow or poor decision making by the project manager.
- ☐ Low performance of the team.

Factors to be considered while selecting the team members in a project.

- ☐ Knowledge about political situation.
- ☐ Knowledge about social situation.
- ☐ Knowledge about economic situation.
- ☐ Experience or skill in developing the strategies.
- ☐ Experience in communication.
- ☐ Experience in fundraising or fund management.
- ☐ Experience in budgeting.
- ☐ Experience in risk calculation.
- ☐ Should understand the psychology of the team.
- ☐ Should not be short tempered.

2.3 Project Design:

- ☐ It is the first stage in the project where ideas are discussed.
- ☐ It involves project scheduling and implementation.
- ☐ It includes finding of location, construction of buildings, buying equipment or machinery and finally finishing of the project.
- ☐ Product design along with the network analysis will help the project manager to develop the work plan to complete the project.

The steps involved in project design are:

Step 1: Identify the total stages in project.

Step 2: Identification of connection between these stages.

Step 3: Developing the control system.

Benefits of Project Design:

- ☐ It helps managers to plan the project economically (careful use of money).
- ☐ It gives an overall idea of the entire project.
- ☐ It shows the time schedule for every project stage (phase).
- ☐ It helps to identify the sequence of events that must occur one-by-one.
- ☐ It is an effective tool for planning and implementation of the project.
- ☐ It is a diagrammatic representation of the work plan to complete the project.
- ☐ It helps the project team to co-ordinate with project activities effectively.

2.4 Work Breakdown Structure (WBS)

- ☐ A work breakdown structure (WBS) is a project management tool that breaks down a project into smaller, more manageable tasks.
- ☐ It is a tree-like structure that shows the different components of a project and how they relate to each other.

Advantages or benefits of WBS.

Here are some of the benefits of using a WBS:

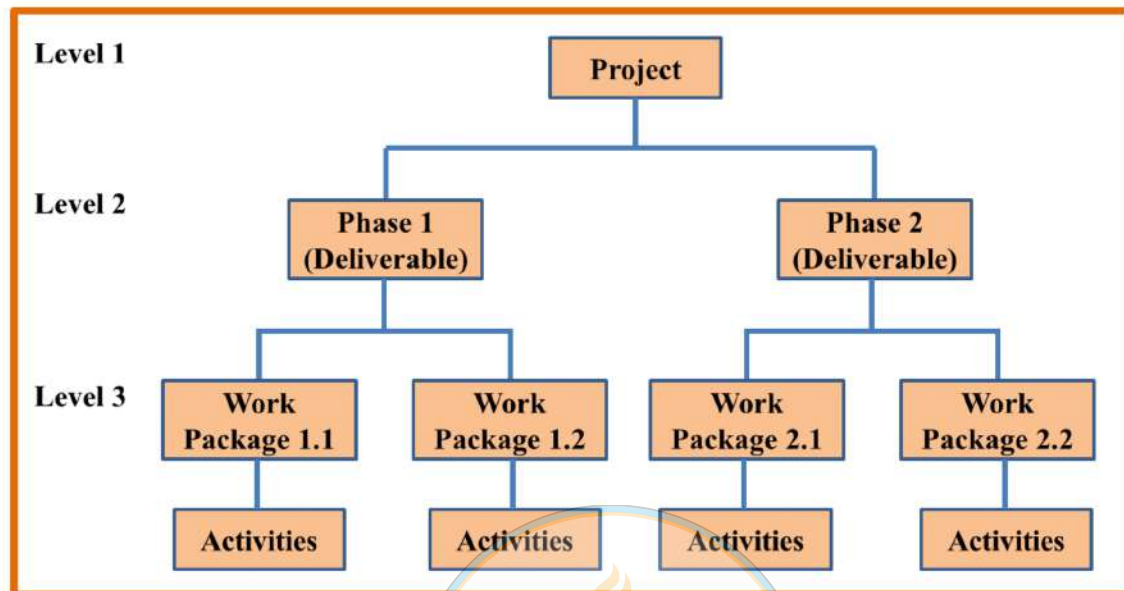
- ☐ It helps to identify all the tasks that need to be completed.
- ☐ It helps to estimate the time, cost, and resources required to complete the project.
- ☐ It helps to identify potential risks and challenges.
- ☐ It helps to track progress and ensure that the project stays on track.
- ☐ It provides a common understanding of the project goals among all stakeholders.

Steps involved in creating a WBS.

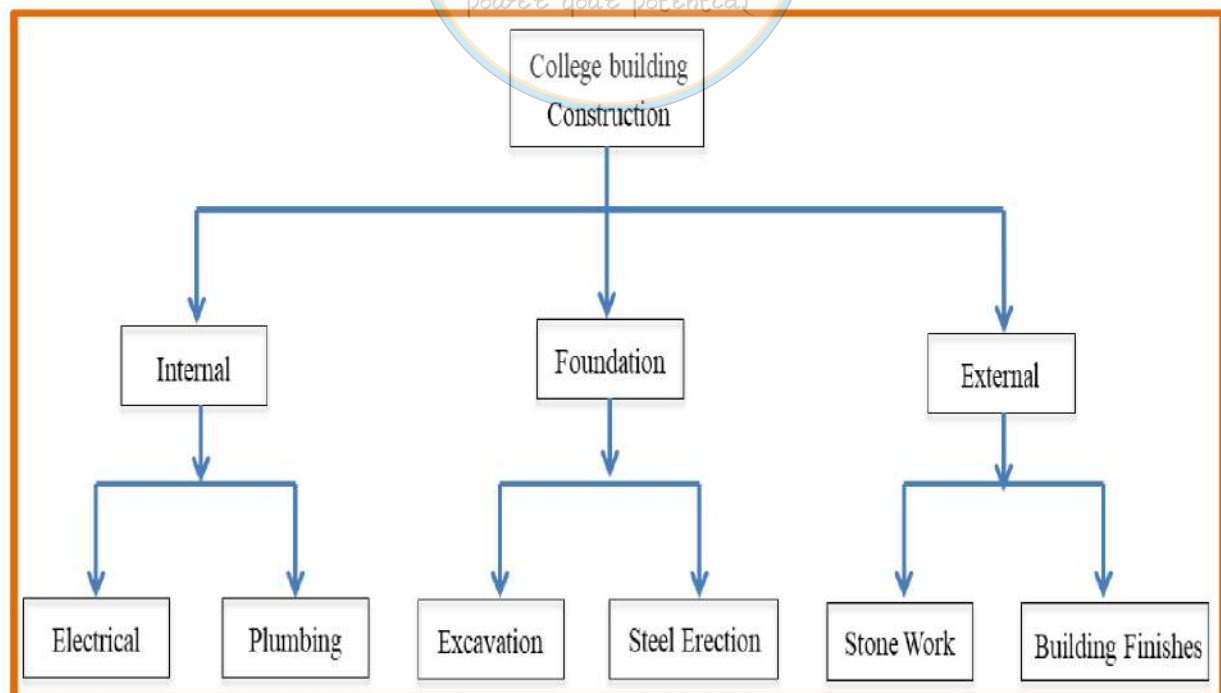
- i. Identify the project's major deliverables (output).
- ii. Break down each deliverable into smaller tasks.
- iii. Continue breaking down the tasks until they are manageable and tasks are assigned to individual team members to estimate time and cost.
- iv. Organize the tasks in tree-like structure where major deliverables at the top and detailed tasks at the bottom.
- v. Label each task with a unique identifier, this will make planning and execution process easy.

Examples for WBS:

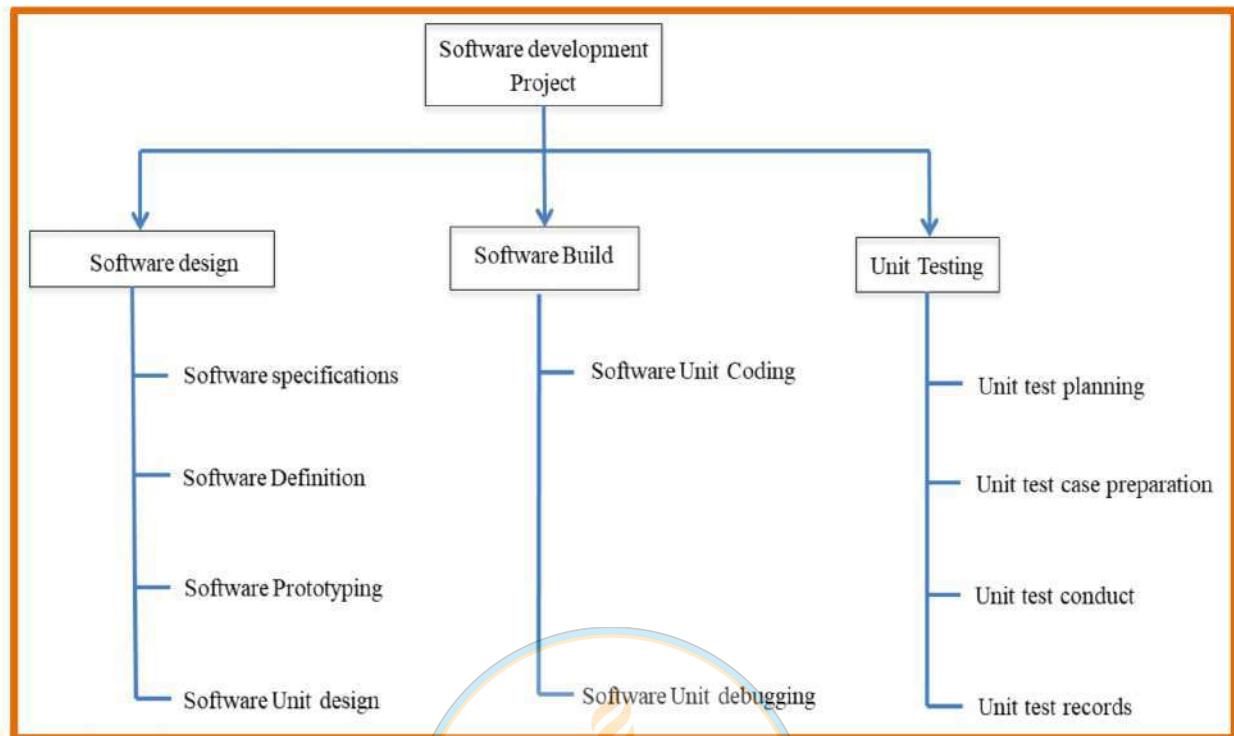
❑ Generalized work breakdown structure (WBS):



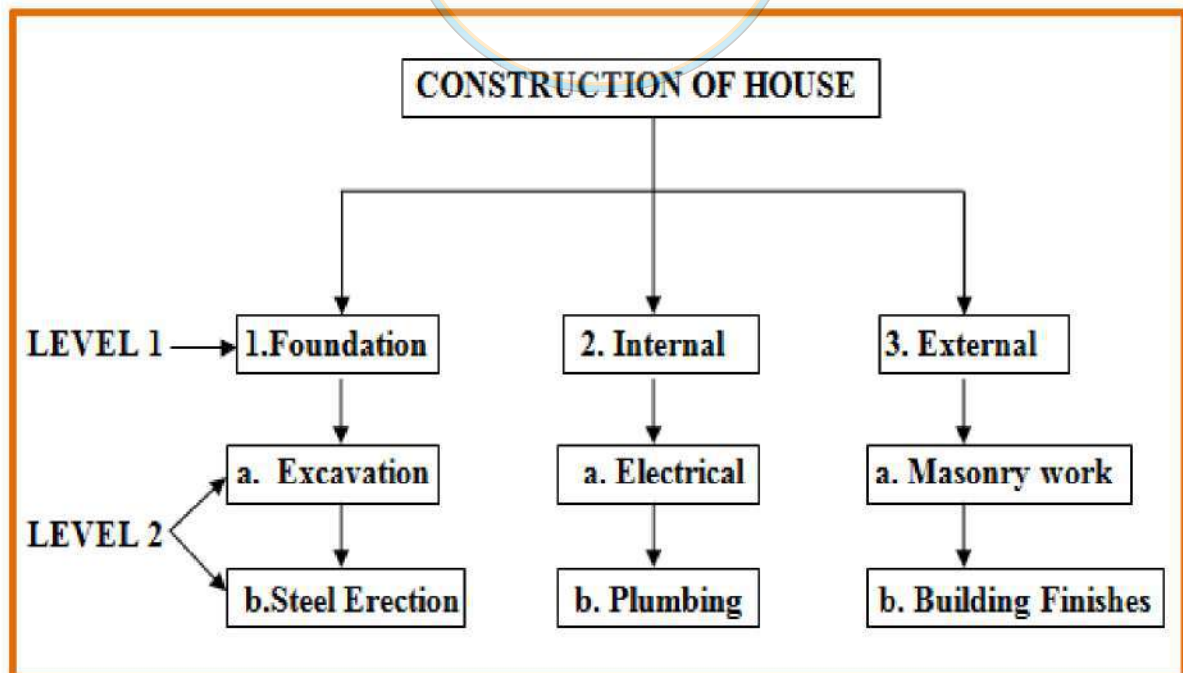
❑ Work Breakdown Structure for the construction of a college building.



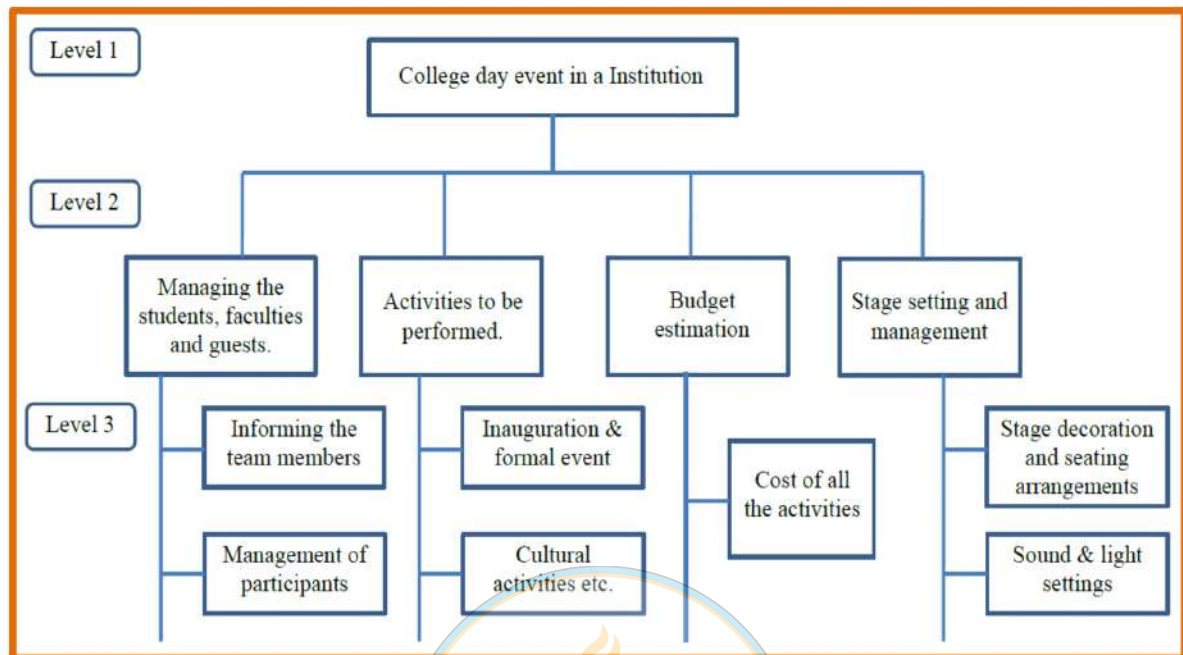
❑ Work Breakdown structure for the software development project.



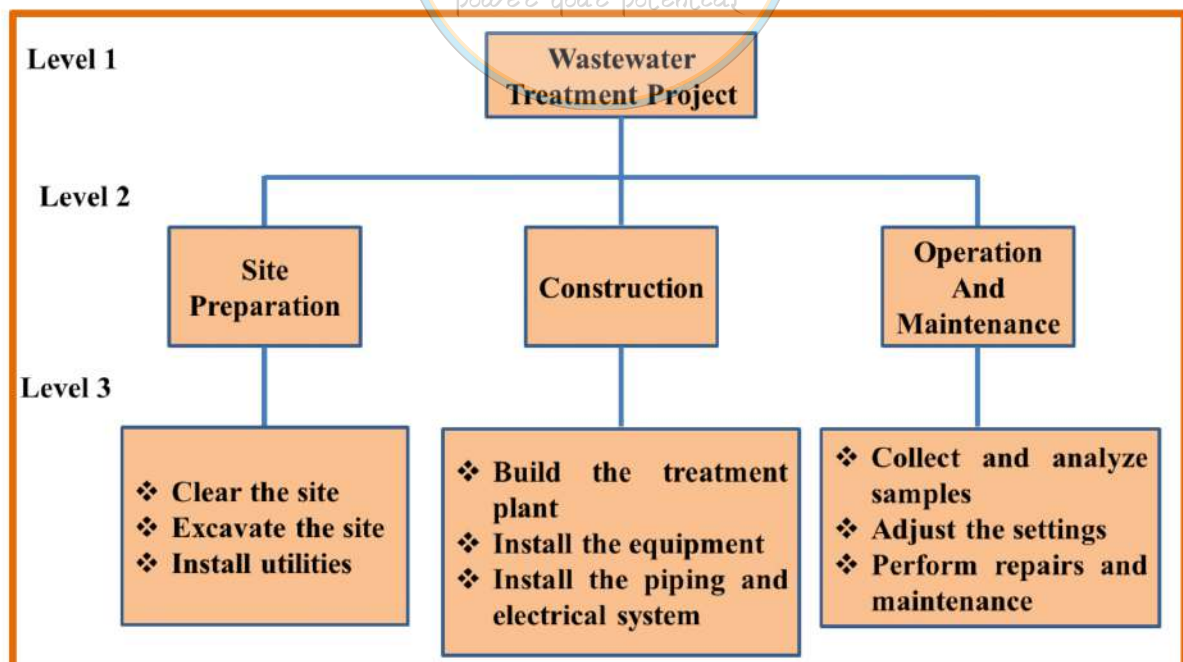
❑ Work Breakdown Structure for the construction of residential house.



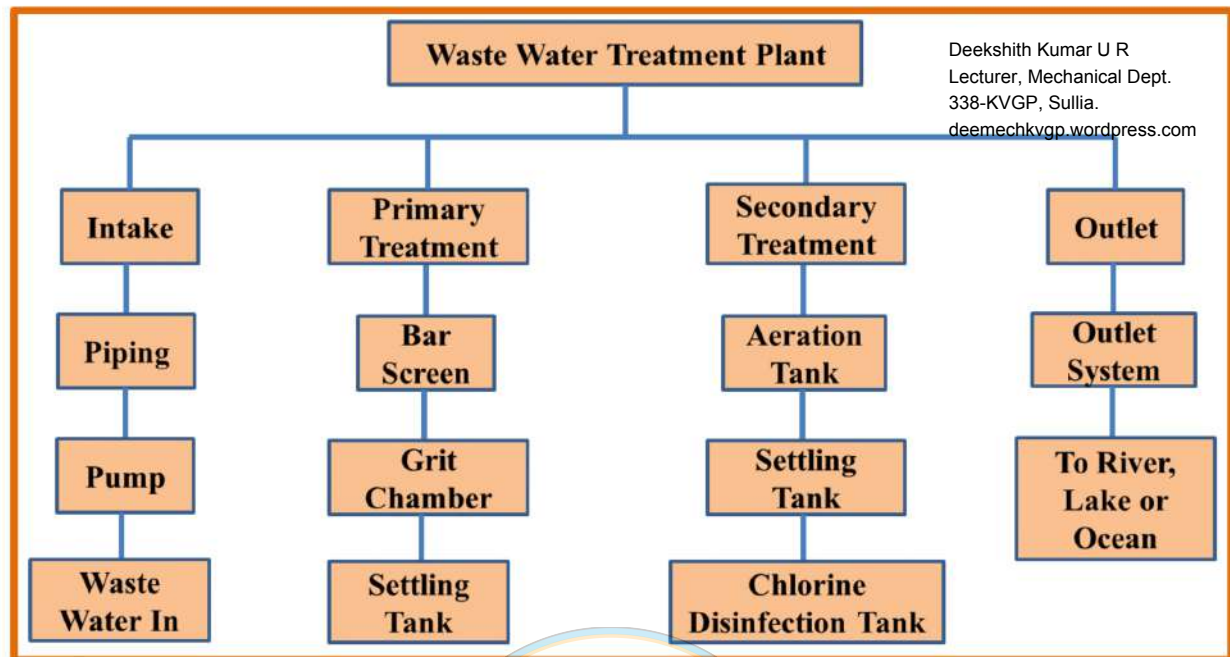
- ❑ Work breakdown structure for conducting college annual day program of your institution.



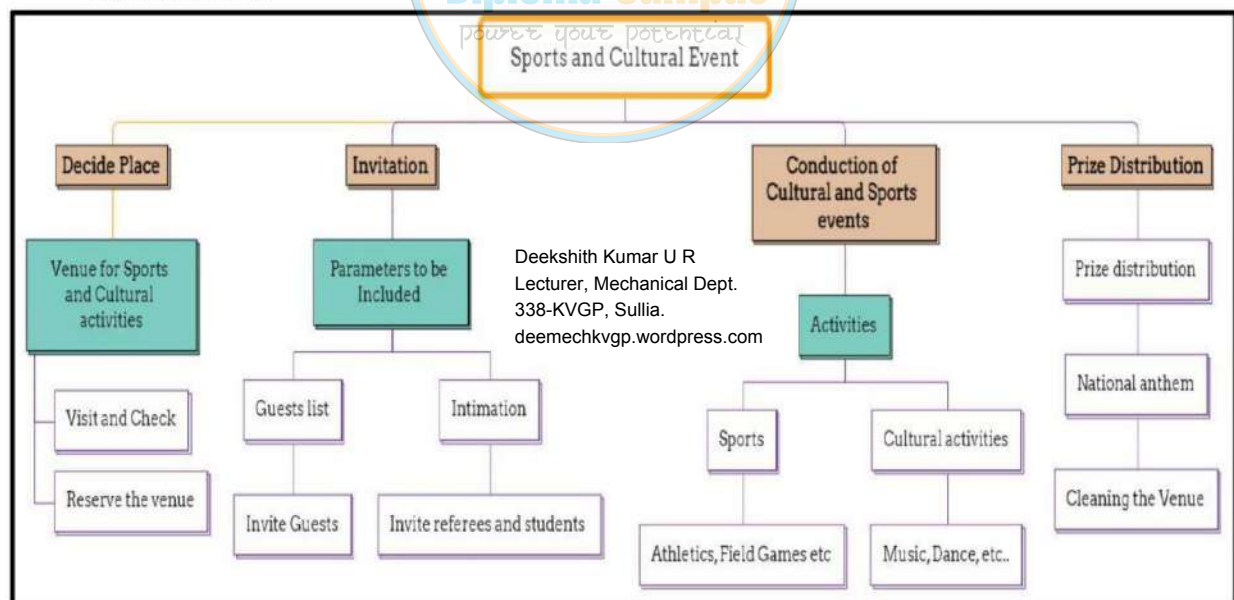
- ❑ Work Breakdown Structure for waste water treatment Project.



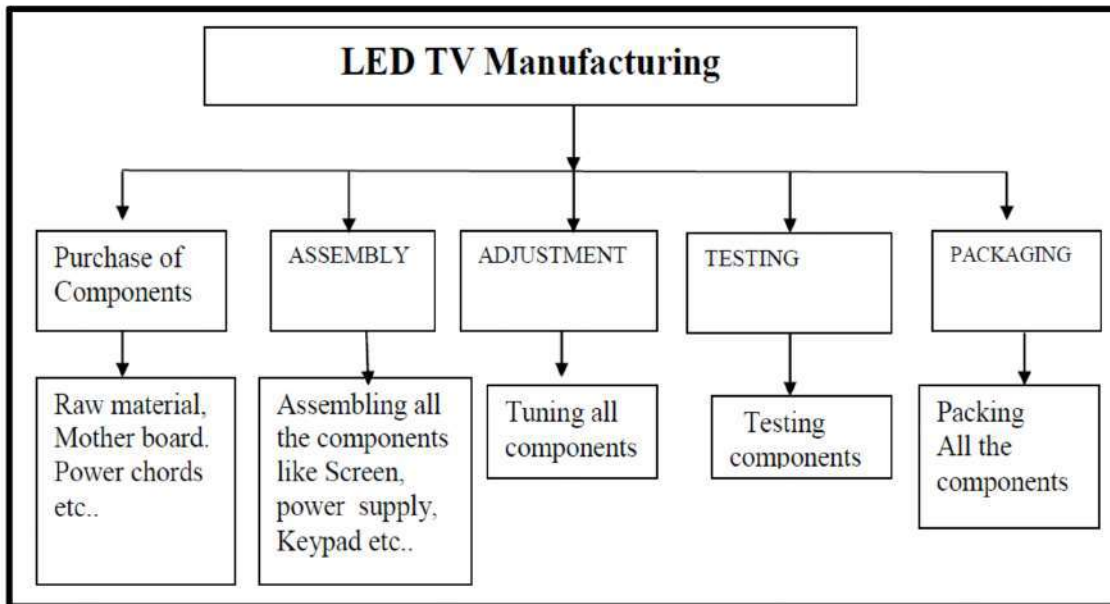
❑ Work Breakdown Structure for Waste Water Treatment Plant.



❑ Work Breakdown Structure of Cultural and Sports Day Event of your Institution.



☐ **Work breakdown structure for manufacturing of LED television.**



2.5 Project Execution Plan (PEP)

- ☐ The Project Execution Plan (PEP) is the main document that explains how a project will be carried out, tracked and managed.
- ☐ It serves as a guide for completing the project successfully.
- ☐ It is a document that describes the objectives (goals), time, resources, costs, quality and benefits of the project.

Key parts of Project Execution Plan (PEP)

- It defines project goals.
- It defines project team roles.
- It helps in identifying task to be completed.
- It helps in resource plan.
- It helps in risk plan.
- It helps in communication plan.
- It helps in time and cost estimation.

Why is it Important?

- ☐ It gives the direction to the project team on what to do.
- ☐ It helps in organizing work and ensures tasks are done in the right order.
- ☐ It avoids problems and risk with proper planning.

PEP includes four sub-plans such as:

- Contracting Plan
- Work packing Plan
- Organization Plan
- Systems and Procedure Plan

2.6 Contracting Plan:

- ☐ A Contracting Plan explains how a project will get the goods, services, raw materials or work from outside suppliers or contractors.
- ☐ It includes who will do the work, how they will be chosen and the terms & conditions of the agreement.

Key Parts of a Contracting Plan

- Lists the tasks or materials required.
- Choose Suppliers (e.g., bids or direct selection).
- Set the cost limits (budget) for contracts.
- Set the deadlines for contracts and need to be signed (when work should be done).
- Define the agreement type (e.g., fixed-price or pay-as-you-go).
- Monitor the contractor work.

Why it is important?

The contracting plan ensures everything needed from outside sources is handled smoothly and fits the project's goals.

- Ensures everyone knows their roles and expectations.
- Save Money
- Reduces problems with suppliers or delays.

2.7 Work Packing Plan

- ☐ A Work Packing Plan is a simple, step-by-step plan that helps manage and organize work for a project.
- ☐ It breaks the project into smaller, manageable parts to make sure the work is done safely, efficiently and correctly.

Key Parts of a Work Packing Plan

- Define, what needs to be done?
- Define, who will do it? (Defines tools or materials they need).
- Define, when it will be done?

- Define safety guidelines to keep everyone safe while working.
- Define, how to do it? (Step-by-step instructions for completing the work).
- Define risks and solutions.
- Track progress.

Why it is important?

A Work Packing Plan helps everyone stay on the same page, work safely and complete the project successfully.

- Helps manage tasks and avoid confusion.
- Ensures Safety and protect workers.
- Saves Time and Effort
- Reduces Risks:

2.8 Organization Plan

- ☐ An Organization Plan is a simple document to show the structure and management of Project organization.
- ☐ It shows the roles and responsibilities of project team.
- ☐ It ensures workflows are clear, making the entire process more efficient.

Key Parts of an Organization Plan

- It defines goals, what project wants to achieve.
- It shows who is in charge (manager) and who does the work (team members).
- It shows the responsibility of each person in the project team.
- It defines the communication plan, how information will be shared among team members to ensure smooth coordination.
- It defines the resource plan (Time, cost, materials, tools etc. required to complete project).
- It defines the rules and regulations, important dates and deadlines.

Why it is important?

- It makes sure people know their tasks.
- It reduces confusion and avoids mistakes.
- It helps in communication so that clear information is shared.
- It keeps the team focused to meet goals.

2.9 Systems and Procedure Plan

- ☐ A Systems and Procedure Plan is a guide that explains how tasks are done within a project by following a set of rules and processes.
- ☐ It ensures work is completed efficiently, consistently and correctly.

Key Parts of a Systems and Procedure Plan

- It provides a clear, instruction on how to complete tasks.
- It defines Roles and Responsibilities of each person.
- It provides lists of equipment, software or materials required to complete tasks.
- It provides important guidelines to ensure quality and safety.
- It defines how to monitor and report work.
- It helps in problem solving.

Why it is important?

- Everyone knows what to do.
- Saves Time, make work faster.
- Ensures Quality.
- Reduces Mistakes.

2.10 Project Procedure Manual (PPM)

- ☐ The project procedure manual gives a complete picture about the project.
- ☐ It is designed to guide project manager.
- ☐ It provides step-by-step instructions, rules, and guidelines to ensure the project runs smoothly from start to finish.

Key Parts of Project Procedure Manual (PPM)

- ☐ It helps project team to follow the same process to avoiding confusion.
- ☐ It explains what to do and how to do it.
- ☐ It clearly defines who is responsible for what.
- ☐ It helps in Quality Management.
- ☐ It helps in Risk Management.
- ☐ It helps in change management.
- ☐ It helps in proper communication.
- ☐ It provides Step-by-step instructions to complete tasks.
- ☐ It helps in officially closing the project and handing over the deliverables to the customers.

Some of the benefits of using a PPM are:

- Increased efficiency
- Reduced risk
- Improved communication
- Improved documentation
- Improved transparency

2.11 Project Diary

- ❑ A project diary is a daily record maintained by Project Manager.
- ❑ Project diary document all key activities, events, decisions, contracts and issues that occur in a project.
- ❑ It acts as a detailed account of the project's progress and serves as a reference for tracking and managing the project effectively.

Importance of Project Dairy:

- A project manager is to conduct number of meetings with vendors (suppliers), contractors, his own team members and various outsiders with respect to the project work.
- It is very difficult to remember everything by human nature.
- Thus Project dairy is used to maintain a record of points discussed and decisions taken in date wise while carrying the project work.

Advantages of Project Dairy:

- This will ensure effective and efficient management.
- This record will justify the decisions in later stage.
- This record is also used to protect against false claims and legal disputes.
- This diary also helps to prepare a follow-up, as this record contains all the pending works with dates committed against each.
- Maintaining the diary helps us to get relief from the burden of carrying everything in our mind.
- It will also boost up our memory power thereby helps us to avoid the problems of unattended work due to lack of memory.

2.12 Project Execution System – List its forms

- ❑ A Project Execution System (PES) is set of processes, tools and methodologies used to complete project from planning to closing.
- ❑ PES helps project managers to plan, track and execute projects.
- ❑ It focus to complete the project within the given time, cost and resources.
- ❑ PES is crucial for managing and controlling the project's resources, schedule and deliverables.

Following are forms of PES:

- **Work breakdown structure (WBS):** Breakdown of the project into smaller and manageable tasks.
- **Gantt charts:** Visual representation of the project plan, showing the tasks, their dependencies, and their start and end dates.
- **Resource management:** Helps to track the resources for each task, such as people, equipment and materials.
- **Cost tracking:** Helps to track the project's budget and expenses and to identify potential cost overruns.
- **Issue tracking:** Helps to track the project's issues, and to resolve them.
- **Communication:** Helps in proper communication between project team members and stakeholders.

2.13 List the Pre-requisites for Successful Project Implementation:

Following are the Pre-requisites:

- i. Adequate formulation (Sufficient Project Formulation).
- ii. Sound project organization.
- iii. Proper implementation planning.
- iv. Advance action.
- v. Timely availability of funds.
- vi. Judicious equipment tendering and procurement.
- vii. Better contract management.
- viii. Effective monitoring.

Meaning of Pre-requisites:

- ☐ Time and cost overruns make the project uneconomical.
- ☐ In order to minimize time and cost over-runs, it is necessary to study about the prerequisites for successful project implementation.
- ☐ Proper use of prerequisites will help to complete the project successfully.

i. Adequate formulation (Sufficient Project Formulation): Problems are easily identified and solved.

ii. Sound project organization: It ensures that the project is well-planned, executed, and controlled

iii. Proper implementation planning: It ensures that the project is executed effectively and desired results are achieved.

iv. Advance action: Quick decision making to solve a problem.

v. Timely availability of funds: Ability to access funds when they are needed.

vi. Judicious equipment tendering and procurement: It ensures purchase of equipment's fits for the need, also ensures transparent and fair price.

vii. Better contract management: Documentation of contract, creating a contract management policy, tracks the status of all contracts and communicates effectively.

viii. Effective monitoring: Track the performance of process, if any problems are identified take quick actions.

2.14 Case Studies

Case Study 1:

Assume you are a member of student association. You are given the responsibilities to organize the college fest.

Requirements:

1. There are 500 students in the college and 50 staffs.
2. There are some activities to be conducted such as Cultural, Sports, and Exhibition etc...
3. Food arrangements for the students and staff.
4. Stage has to be setup.
5. Prize distribution on college fest day.

Tasks:

- Split the class into groups of three or five.
- Prepare the detailed WBS.
- Prepare the detailed Project execution plan.
- Estimate the budget for the entire activity.
- Present it in a seminar, with each group getting 5-10 minutes to present their idea.

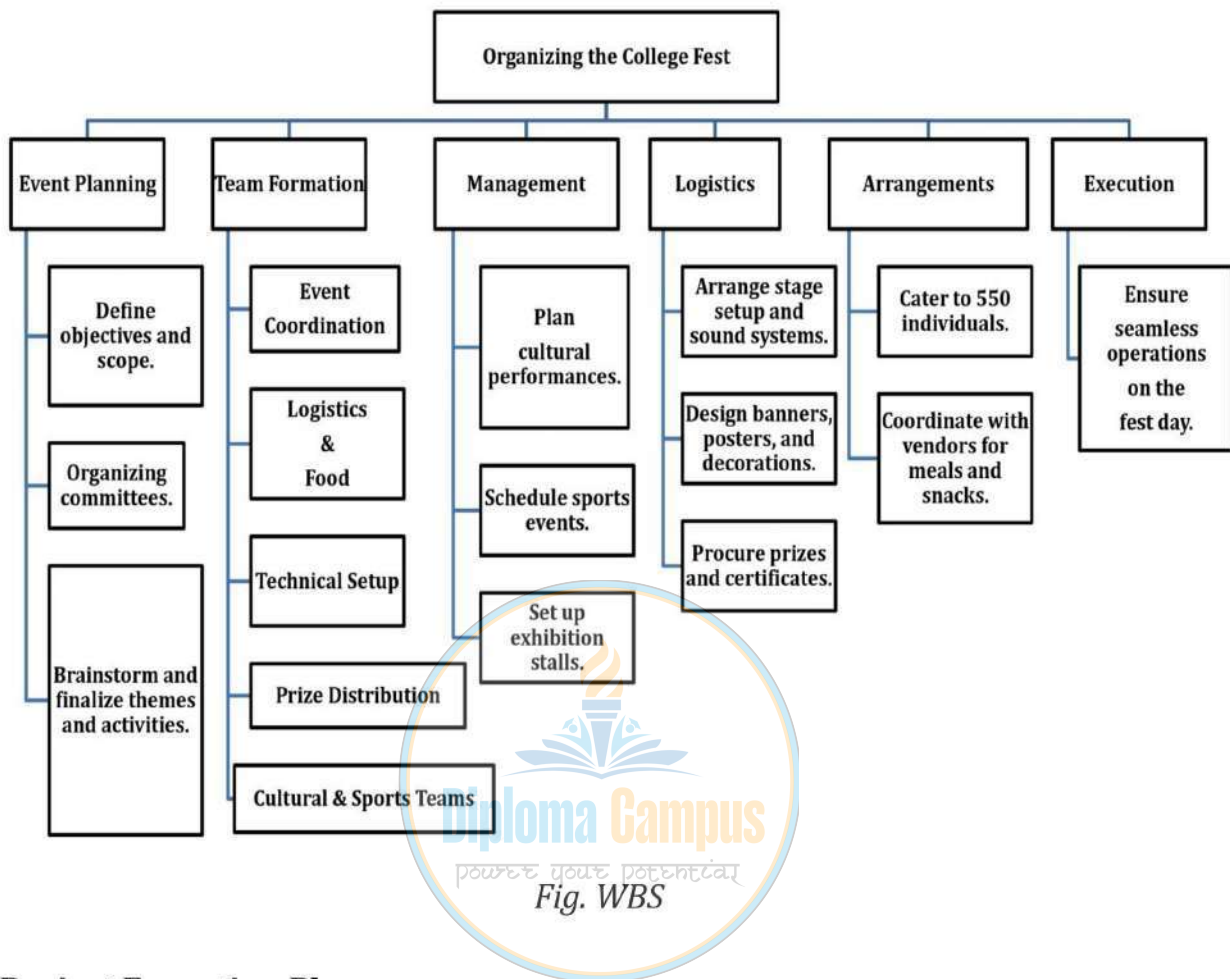
Case Study 1 Solution:

Case Study: Organizing the College Fest

Objective: Organize and execute a successful college fest for 500 students and 50 staff members, incorporating cultural, sports, and exhibition activities, along with food arrangements, stage setup, and prize distribution.

Work Breakdown Structure (WBS):

The project can be divided into the following major components:



Project Execution Plan:

1. Pre-Planning Phase:

- Form teams and assign responsibilities.
- Prepare a detailed schedule with timelines.
- Get approvals from the administration.
- Invite chief guests or dignitaries.

2. Execution Phase:

- Cultural Team: Manage auditions, rehearsals, and event execution.
- Sports Team: Prepare grounds, finalize equipment, and manage matches.
- Logistics Team: Coordinate with decorators and vendors for smooth setup.
- Food Team: Ensure timely delivery and quality of food.

3. Post-Event Phase:

- Distribute prizes and mementos.
- Gather feedback for improvements.

Estimated Budget:

Expense Category	Cost (INR)
Stage Setup (Sound, Lights, Decor)	30,000
Food (550 individuals @ INR 200/person)	1,10,000
Prizes & Certificates	20,000
Banners & Marketing	10,000
Cultural & Sports Equipment	15,000
Miscellaneous	15,000
Total	2,00,000

Seminar Presentation Plan:

- 1. Introduction** (1 minute)
 - Objective of the event.
 - Importance of teamwork.
- 2. WBS Explanation** (2-3 minutes)
 - Visualize WBS through charts.
- 3. Project Execution Plan** (2-3 minutes)
 - Explain phases with key milestones.
- 4. Budget Analysis** (2 minutes)
 - Highlight major expenses.
- 5. Q&A Session** (1 minute)
 - Address queries from the audience.

Conclusion: This structure ensures systematic planning and execution, promoting collaboration and skill-building among students.

Question bank (05 to 10 Marks):

1. Explain Essentials of Project Administration.
2. Define project administration and list any 3 tools used in project administration.
3. Define project team and list the types of project team.
4. Explain different types of project team.
5. What are the advantages of effective team?
6. List the disadvantages of ineffective team.
7. What are the factors considered while selecting project team members?
8. Explain Project Design.
9. List the benefits of project team.
10. Explain WBS.
11. List the benefits of WBS.
12. What are the steps involved in creating WBS?
13. Develop a generalized WBS.
14. Develop WBS for construction of college building.
15. Develop WBS for construction of house.
16. Develop WBS for software development project.
17. Develop WBS for conducting college annual day program of your institution.
18. Develop WBS for waste water treatment Project.
19. Develop WBS for waste water treatment Plant.
20. Develop WBS of Cultural and Sports Day Event of your Institution.
21. Develop WBS for manufacturing of LED television.
22. Explain PEP.
23. Explain Contracting Plan.
24. Explain Work Packing Plan.
25. Explain Organization Plan.
26. Explain A Systems and Procedure Plan.
27. Explain PPM.
28. Explain Project dairy.
29. What are the advantages of project dairy?
30. Explain Project Execution System (PES).
31. List the forms in PES.
32. List the Pre-requisites for Successful Project Implementation.

